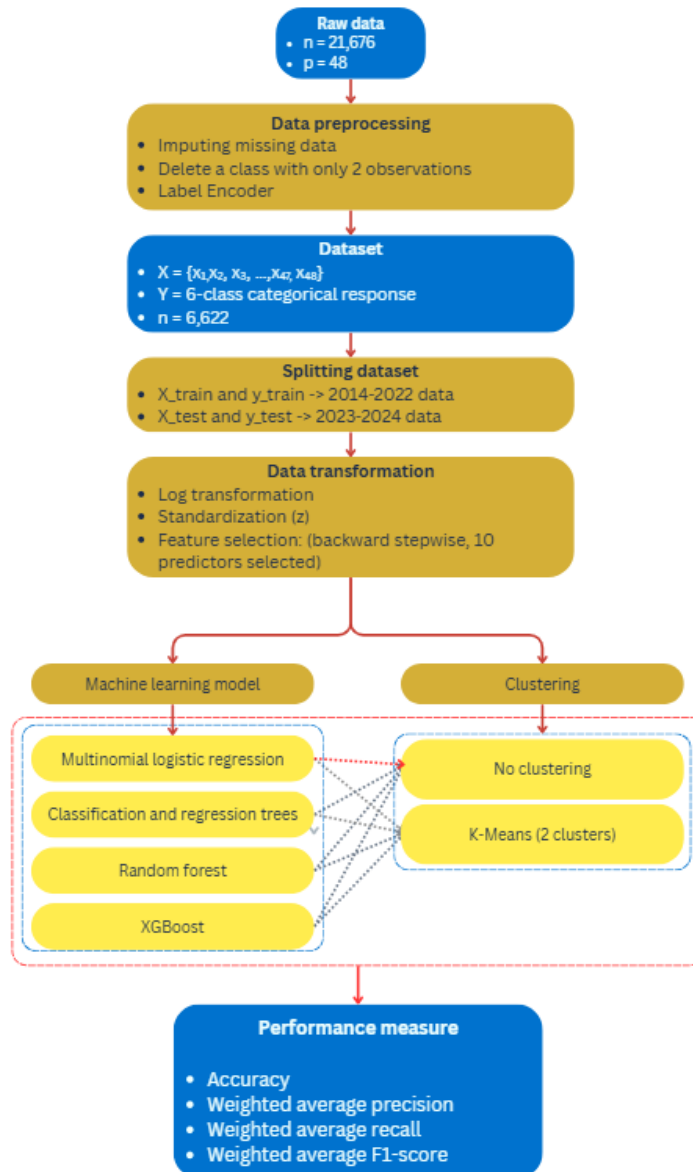


Graphical Abstract

Data-driven corporate credit assessment: Machine learning approaches to credit rating prediction

Mensel Santoso, Wu Lung Chieh, Frank Hsu, Huei-Wen Teng



Highlights

Data-driven corporate credit assessment: Machine learning approaches to credit rating prediction

Mensel Santoso, Wu Lung Chieh, Frank Hsu, Huei-Wen Teng

- This paper implements hybrid machine learning (clustering and classification) to improve the accuracy of credit rating models.
- The features are first scaled, followed by clustering to form more homogeneous subgroups. Classification is then conducted within each subgroup to enhance predictive precision.
- Comparisons show that this hybrid machine learning gives better prediction than standalone machine learning method.

Data-driven corporate credit assessment: Machine learning approaches to credit rating prediction

Mensel Santoso, Wu Lung Chieh, Frank Hsu, Huei-Wen Teng

National Yang Ming Chiao Tung University, No 1001 Daxue Road East District, Hsinchu City, 300093, Taiwan, R.O.C.

Abstract

Credit rating prediction is a cornerstone of financial risk assessment, yet traditional statistical models often fall short in capturing the nonlinearities and heterogeneities inherent in corporate financial data. This study proposes a hybrid machine learning framework that integrates clustering and classification techniques to enhance both predictive accuracy and interpretability. Specifically, we explore combinations of K-means clustering with multinomial logistic regression, classification and regression trees (CART), random forest, and XGBoost. Clustering is used to segment heterogeneous financial data into more homogeneous subgroups, allowing classifiers to learn more precise decision boundaries within each cluster. Using Taiwan Corporate Rating Index dataset, we evaluate multiple hybrid configurations and benchmark them against standalone models. Our results demonstrate that hybrid architectures significantly improve default prediction accuracy, reduce misclassification rates, and offer practical advantages in handling unrepresentative samples and imbalanced classes. This research contributes to the development of adaptive, interpretable, and high-performing credit rating systems for modern data-driven financial environments.

Keywords: Credit Rating, Hybrid Machine Learning, Clustering, Classification

JEL: C38, C53, G21, G32

1. Introduction

Credit ratings play a pivotal role in financial markets, influencing investment decisions, regulatory capital requirements, and corporate financing costs (Afik and Galil, 2024). As the complexity of financial systems grows, so does the need for predictive models that are both accurate and interpretable. Traditional statistical approaches, such as logistic regression and discriminant analysis, have long been used for credit scoring (Galil et al., 2023; Hajek et al., 2024), but they often struggle with nonlinear patterns, imbalanced datasets, and unrepresentative samples.

Recent advances in machine learning have opened new avenues for credit rating prediction. Neural networks, support vector machines, and ensemble methods have demonstrated superior performance in capturing complex relationships (Wu et al., 2014; Galil et al., 2023). However, these models often lack transparency, a critical requirement in regulated financial environments (Teng et al., 2024). To address this, hybrid machine learning models, which combine clustering and classification, have emerged as a promising solution.

Hybrid models leverage the strengths of unsupervised clustering to preprocess data, identify latent structures, and isolate noisy or inconsistent samples (Tsai and Chen, 2010; Hsieh, 2005). These clusters can then be used to train more focused classifiers, improving both accuracy and interpretability. For instance, Hsieh (2005) demonstrates that the combination of self-organizing maps with neural networks enhances the performance of credit scoring by filtering out unrepresentative samples. Similarly, Tsai and Chen (2010) show that clustering-based preprocessing improves the robustness and precision of credit scoring prediction.

Moreover, Galil et al. (2023) emphasize the importance of monotonicity in credit scoring models, advocating for interpretable CART-based (Classification and Regression Tree) architectures that align with financial theory. Teng et al. (2024) introduce a rescaled cluster-then-predict method that balances predictive power and interpretability by combining XGBoost and logistic regression within clustered subgroups. These approaches reflect a broader trend toward explainable AI in finance, driven by regulatory mandates such as GDPR and the Fair Credit Reporting Act.

In this study, we build on these insights to develop a hybrid machine learning framework for predicting corporate credit ratings. Using Taiwans TCRI dataset, we explore multiple hybrid configurations, including clustering-then-classification pipelines, and benchmark them against standalone models.

Our goal is to assess not only predictive accuracy but also model transparency, high interpretability, robustness to imbalanced data, and alignment with financial theory.

By integrating a clustering technique, in this case K-means, with classifiers such as multinomial logistic regression, CART, random forest, and XGBoost, we aim to contribute to the development of adaptive, interpretable, and high-performance credit scoring systems. This research aligns with the growing demand for data-driven and explainable financial models that offer practical implications for banks, regulators, and investors.

2. Data

2.1. Predictors and response variable

The dataset employed in this study comprises financial and accounting indicators of publicly listed firms in Taiwan, sourced from the Taiwan Economic Journal (TEJ). The response variable is the Taiwan Corporate Credit Risk Index (TCRI), an ordinal categorical measure of firm-level credit ratings ranging from 5 to 9 and D. In its raw form, the dataset contains 48 predictors and a multiclass response variable. For modeling purposes, a subset of ten predictors was retained as the final feature set, as summarized in Table 1, while the complete list of predictors and their data types is provided in Appendix A. The following subsections present data exploration, missing values, outliers, and descriptive statistics to establish the characteristics of the dataset prior to modeling.

Table 1: Selected financial predictors

No.	Variable	Type
1	total liabilities	Continuous
2	share capital	Continuous
3	total capital	Continuous
4	finance costs	Continuous
5	current ratio	Continuous
6	quick ratio	Continuous
7	interest expense ratio (B)	Continuous
8	total liabilities/total net worth	Continuous
9	operating profit/paid-in capital ratio	Continuous

No.	Variable	Type
10	retention ratio	Continuous

2.2. Data exploration

Exploration of this dataset focused on identifying missing values, assessing the presence of outliers, and summarizing the distributional properties of the predictors and response variable. Descriptive statistics and visualizations were employed to highlight variability across firms and to illustrate how financial indicators align with observed TCRI grades. The following subsections present these findings in detail, providing the empirical foundation for the modeling procedures introduced in Section 3.

2.2.1. Missing values

To ensure data integrity and modeling reliability, missing values were systematically assessed across all predictors. Table 2 summarizes the extent of missingness in key predictors. The most affected predictor, expected credit impairment (loss) profit – operating expense, exhibited a missing rate of 36.79%, followed by retention ratio (21.97%) and operating leverage (14.43%). As later shown in the feature selections, most of these variables are not selected by the backward stepwise feature selections even though all the missing values were imputed using median as their values.

Table 2: Summary of missing values in predictors

Variable	Missing Values	Missing (%)
Expected credit impairment (loss) profit – operating expense	2,437	36.80
Retention ratio	1,455	21.97
Operating leverage	956	14.43
Financial leverage	845	12.76
R&D expense	232	3.50
Interest coverage ratio	191	2.88
Cash flow adequacy ratio	107	1.62
Cash reinvestment ratio	36	0.54
Contingent liabilities to net worth	2	0.03
ROE – comprehensive income	1	0.02
ROA – comprehensive income	1	0.02

2.2.2. Outliers

Outliers are identified throughout the dataset using the standard threshold of $|z\text{-score}| > 3$ in each column, which flags values that deviate substantially from the mean. This pattern reflects the natural variability in financial metrics across firms and industries, where size, structure, and performance can differ widely. Rather than excluding these observations, we retain all data points to preserve the full financial landscape. Log transformation and scaling are applied to reduce the influence of extreme values on model estimation. A complete set of box plots and outlier counts is provided in Appendix B to illustrate the distributional characteristics and prevalence of outliers across predictors.

2.2.3. Descriptive statistics of predictors and response variable

Table 3 reports the descriptive statistics of the ten predictors retained after log transformation and standardization. The summary reflects the original values prior to scaling, providing a view of the raw distributional properties across 6,622 firmyear observations.

Table 3: Descriptive statistics of selected predictors

Variable	Count	Mean	Std. Dev.	Min	25%	Median	75%	Max	Skewness	Kurtosis
Total liabilities	6622	11552982.53	62382019.86	4946.00	944673.75	2295347.00	5886414.00	1607622209.00	15.824	307.617
Share capital	6622	3400133.46	7710776.71	157600.00	875460.00	1490550.00	3005579.00	105596201.00	7.849	78.689
Total capital	6622	1191068.29	5305519.35	-207088.00	59116.25	305537.50	879875.25	105919701.00	14.138	236.684
Finance costs	6622	109532.60	444821.87	0.00	4889.25	18172.50	57624.00	9556030.00	10.997	157.611
Current ratio	6622	359.3196	1518.8762	5.88	135.6825	183.7050	285.4175	71149.48	27.090	973.155
Quick ratio	6622	238.5410	789.1300	0.43	68.2750	114.3650	197.5925	18744.78	14.626	262.141
Interest expense ratio (B)	6622	3.6716	77.2482	0.00	0.16	0.56	1.36	5851.14	67.100	4980.851
Total liabilities / total net worth	6622	120.8301	543.0664	0.48	42.7750	80.3150	134.2325	41648.92	68.020	5168.925
Operating profit / paid-in capital ratio	6622	17.1705	40.4674	-777.80	-1.21	9.09	28.0125	631.99	2.509	58.180
Retention ratio	6622	15.4171	776.7598	-59557.14	21.8550	37.2850	54.1875	100.00	-69.230	5240.150

Several predictors exhibit extreme skewness and kurtosis. For instance, the interest expense ratio and total liabilities to net worth display heavy tails, indicating that values are spread far beyond the central range. Measures such as the current ratio and quick ratio reveal wide dispersion, with extreme outliers dominating the distribution. In contrast, operating profit / paid-in capital ratio is more moderately skewed. These distributional features highlight the importance of applying log transformations and scaling to reduce the dominance of tails and to harmonize units across all predictors.

The selected predictors are directly linked to the Taiwan Corporate Credit Risk Index (TCRI), which grades firms from 1 (highest credit quality) to D (default). In this dataset, ratings range only from 5 to 9 and D, reflecting firms with moderate to high risk profiles. For more comprehensive review, the descriptive statistics for all the predictors is attached as Appendix C.

Table 4: Distribution of TCRI ratings

TCRI Rating	Count	Percentage (%)
5	2054	31.01
6	2161	32.62
7	1389	20.97
8	667	10.07
9	292	4.41
D	59	0.89
C	2	0.03

2.2.4. Data visualization: Scatter plot

The pairwise scatter plot of the ten selected predictors (in original, untransformed form) reveals varied relationships across firm size, profitability, and leverage indicators, with nonlinear patterns and visible outliers, particularly in interest expense and operating profit, supporting the earlier decision to apply log transformation and standardization. These predictors span key financial dimensions relevant to credit risk, and their distributional spread aligns with TCRI grades, where firms with stronger profitability tend to cluster at higher ratings, while extreme leverage and liabilities are more common among lower-rated or defaulted firms.

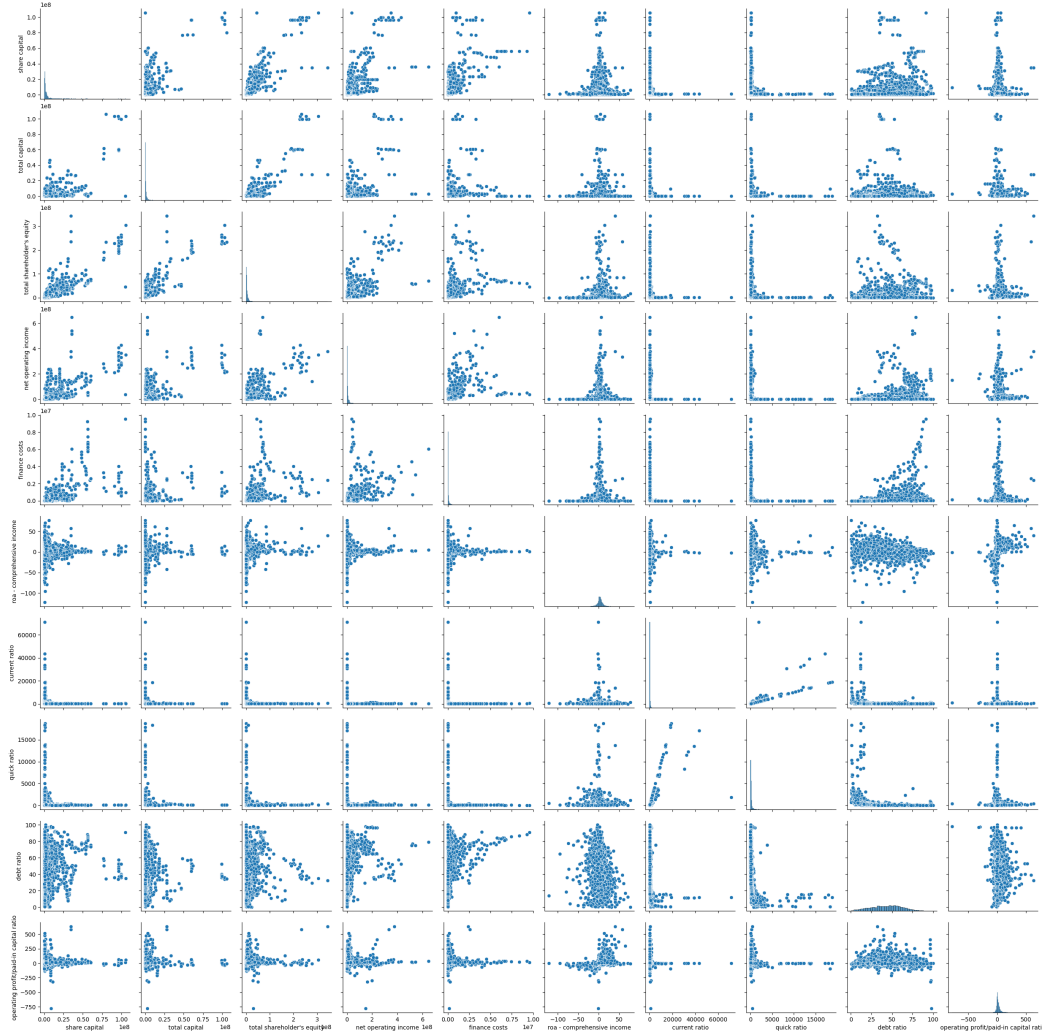


Figure 1: Pairwise scatter plot of selected predictors

2.2.5. Data visualization: Heatmap

The correlation heatmap of the ten selected predictors, based on original pretransformation values, highlights the statistical relationships among the variables. Size-related measures such as share capital, total capital, and finance costs show moderate to strong positive correlations, with share capital and total capital at 0.70 and share capital and finance costs at 0.69. Liquidity indicators, current ratio and quick ratio, are highly correlated at 0.74. Other variables, including interest expense ratio (B), total liabilities to net

worth, operating profit to paid-in capital ratio, and retention ratio, exhibit weak or near-zero correlations with most predictors. Overall, the heatmap demonstrates that while certain groups of variables cluster together with strong associations, many predictors remain largely independent, providing complementary statistical dimensions for subsequent modeling.

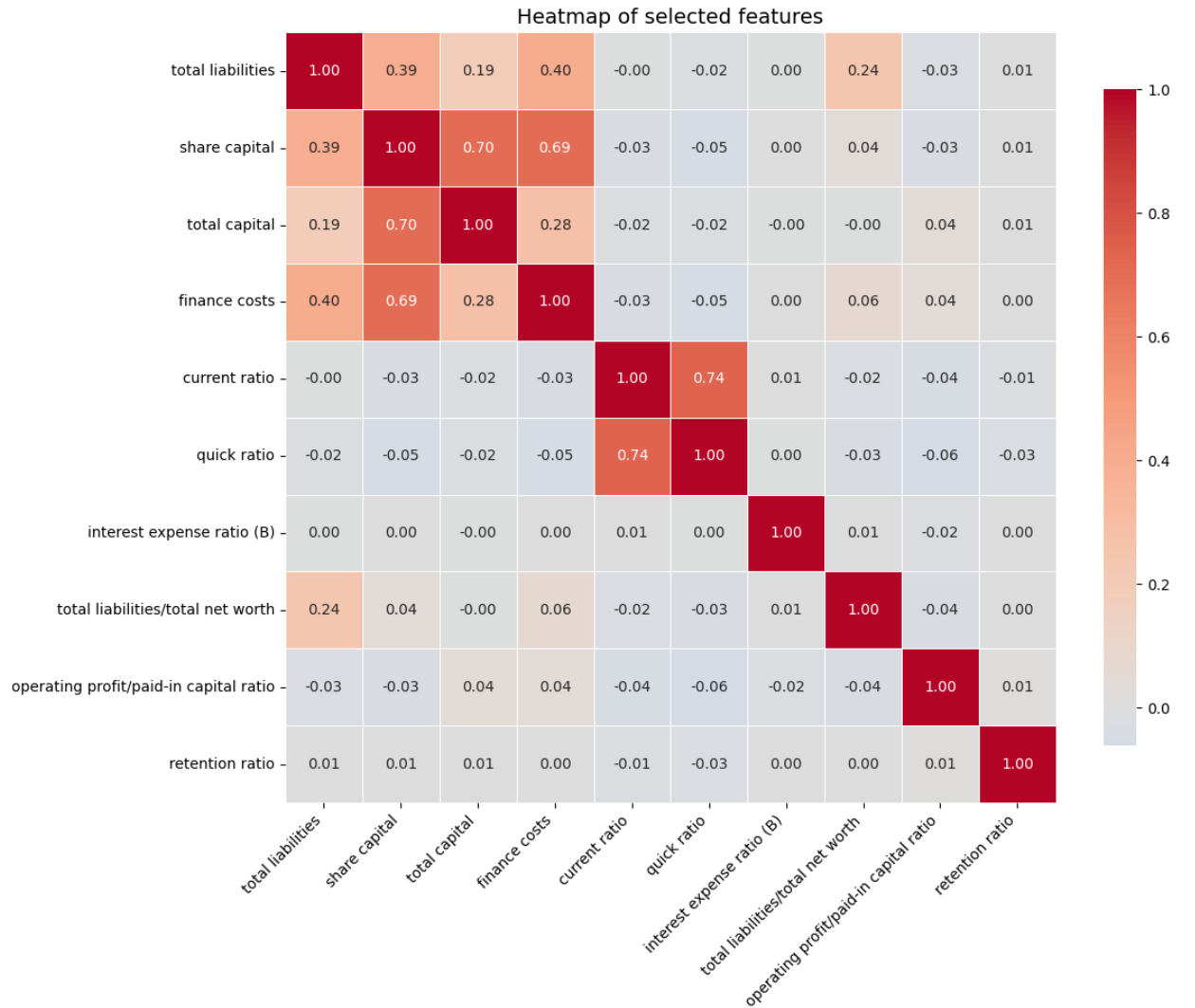


Figure 2: Heatmap of selected predictors

3. Methodology

3.1. Benchmark model

To establish a reliable baseline for model comparison, this study employs multinomial logistic regression (MLR) as the benchmark model for predicting corporate credit ratings (TCRI). Logistic regression has been one of the most extensively used statistical approaches in credit scoring and default risk modeling due to its interpretability, theoretical grounding, and regulatory acceptance. By modeling the log-odds of categorical outcomes as a linear combination of explanatory variables, MLR enables straightforward interpretation of the marginal effect of each financial ratio or firm attribute on the probability of belonging to a particular credit rating category.

In this research, the multinomial extension of logistic regression is applied because TCRI represents a categorical rating system with multiple ordered levels, rather than a binary classification. The model estimates the probability of each firm-year observation being assigned to one of several TCRI classes, using financial indicators derived from firms balance sheets, income statements, and ESG-based scores as predictors. These include liquidity ratios, leverage ratios, profitability measures, and capital structure variables, all of which are theoretically and empirically linked to creditworthiness.

Multinomial logistic regression serves as a baseline for two primary reasons. First, it provides an interpretable benchmark that can be used to evaluate the incremental predictive gains achieved by more complex models such as CART and XGBoost (Barboza et al., 2017). Unlike black-box methods, the coefficients of MLR can be directly interpreted in terms of odds ratios, offering transparency and alignment with financial theory (Guidotti et al., 2018).

Second, MLR is robust to moderate data imbalance and is well-suited to identify monotonic relationships between financial variables and credit risk, which are fundamental to the principles of credit evaluation.

Although logistic regression may be limited in capturing nonlinear interactions and latent structures inherent in financial data, its simplicity and interpretability make it an ideal starting point for benchmarking. The model thus provides a reference level of performance against which the effectiveness of hybrid and machine learning-based models can be rigorously compared in subsequent analyses.

3.2. Design of Research

This study adopts a modular, hybrid machine learning framework to enhance the predictive accuracy and interpretability of corporate credit rating models. The research design is structured around three core phases: data preprocessing, unsupervised clustering, and supervised classification, each chosen to address specific limitations in traditional credit scoring approaches.

3.2.1. Phase 1: Data Preprocessing and Feature Selection

The raw dataset consists of 21,767 bi-annual firm-year observations spanning 2014 to 2024, sourced from the Taiwan Economic Journal (TEJ). Each record includes 48 financial and accounting predictors and one response variable (TCRI). Since TEJ reports both midyear and endofyear values, midyear entries were excluded, resulting in 11,014 endofyear observations. TEJ coverage exceeds that of TCRI, and 4,390 records lacked valid TCRI ratings. To ensure model stability, one rating category with only two observations was removed, yielding 6,622 complete observations. We then cleaned and transformed the TCRI dataset by imputing missing values with medians, applying log transformations to highly skewed variables to normalize distributions and reduce the influence of outliers, and conducting feature selection guided by financial theory and correlation analysis to retain the most relevant predictors of credit risk.

3.2.2. Phase 2: Clustering for Structural Segmentation

To address data heterogeneity and improve model focus, we apply unsupervised clustering techniques, i.e., k-means, to segment firms into more homogeneous subgroups. We expect these clusters to capture latent structures in the financial data, such as firm size, leverage profiles, or profitability patterns. By isolating structurally similar firms, we reduce noise and allow classifiers to learn more precise decision boundaries. This cluster-then-predict strategy is designed to overcome the limitations of global models that struggle with unrepresentative samples and class imbalance.

Table 5: Performance of Multinomial Logistic Regression under different feature selection strategies

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
MLR: All predictors	0.55	0.54	0.55	0.54
MLR: Forward 10 preds	0.51	0.51	0.51	0.50
MLR: Forward 20 preds	0.52	0.51	0.52	0.51
MLR: Forward auto	0.52	0.51	0.52	0.51
MLR: Backward 10 preds	0.55	0.54	0.55	0.54
MLR: Backward 20 preds	0.54	0.53	0.54	0.53
MLR: Backward auto	0.54	0.54	0.54	0.53

Feature selection was subsequently performed to reduce redundancy and retain the most informative predictors of credit risk. Both forward and backward selection procedures were applied, with fixed predictor counts (10 and 20) as well as automatic stopping rules. The results showed that backward selection provided slightly stronger performance (accuracy up to 0.55 with 10 predictors) compared to forward selection (accuracy around 0.51-0.52). Notably, the multinomial logistic regression model using all predictors achieved the same performance measures as the backward stepwise model with 10 selected features (accuracy 0.55, precision 0.54, recall 0.55, F1-score 0.54). Weighted precision, recall, and F1-scores followed similar patterns, confirming that backward selection yields a marginal advantage in balancing model fit and parsimony while reducing the predictor set without sacrificing accuracy.

3.2.3. Phase 3: Classification within Clusters

Within each cluster, we train supervised classifiers, i.e., multinomial logistic regression, CART, Random Forest, and XGBoost to predict TCRI ratings. This localized modeling approach improves both accuracy and interpretability. Logistic regression provides a transparent benchmark, CART offers rule-based insights aligned with financial monotonicity, Random Forest leverages ensemble averaging to reduce variance and highlight stable predictor importance, and XGBoost captures nonlinear interactions with high predictive power. Model performance is evaluated using cross-validation and test-set accuracy, with additional metrics such as precision, recall, and F1-score to assess robustness.

3.2.4. Comparative Benchmarking

To validate the effectiveness of the hybrid framework, we benchmark each cluster-based model against its standalone counterpart trained on the full dataset. This comparison quantifies the incremental gains in predictive accuracy and interpretability achieved through clustering. We also assess the models sensitivity to class imbalance and their ability to generalize across different firm profiles.

3.2.5. Research Contribution

This research design integrates structural segmentation with predictive modeling to construct a scalable and interpretable framework for credit rating prediction. The approach is motivated by the need to balance statistical rigor with the flexibility of modern machine learning methods. By embedding clustering and feature selection into the preprocessing phase, the design reduces heterogeneity and ensures that predictors reflect complementary dimensions of firm performance. This methodological structure not only supports robust classification but also aligns with the growing demand for explainable AI in finance. The framework is intended to be pedagogically valuable for academic research and practically relevant for regulators and financial institutions, as it demonstrates how transparent modeling choices can be combined with advanced algorithms to yield interpretable and highperformance credit scoring systems.

3.3. Performance measures

Given that the response variable (TCRI) is a multi-class categorical rating system, model evaluation requires metrics that capture both overall accuracy and class-specific performance. We therefore report four complementary measures: accuracy, weighted precision, weighted recall, and weighted F1-score. Accuracy provides a global measure of correct classifications, while precision, recall, and F1-score are computed as weighted macro averages, accounting for class imbalance and highlight the trade-offs between false positives and false negatives across rating categories.

These metrics are applied consistently to the benchmark multinomial logistic regression model and to the proposed hybrid framework that integrates clustering with classification (K-means combined with logistic regression, CART, random forest, and XGBoost). This comparative evaluation allows us to quantify the incremental gains in predictive accuracy, robustness, and

interpretability achieved through structural segmentation, thereby assessing the practical value of the cluster-then-predict strategy in credit risk modeling.

4. Results

The empirical evaluation compares the benchmark multinomial logistic regression model with alternative classifiers (CART, Random Forest, and XGBoost), both in standalone form and within the proposed cluster-then-predict framework. Performance is assessed using accuracy, weighted precision, weighted recall, and weighted F1-score, consistent with the multi-class nature of the TCRI response.

Table 6: Performance comparison of models with and without clustering

Group	Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
No cluster	Multinomial Logistic Regression	0.54	0.54	0.54	0.53
No cluster	CART	0.48	0.48	0.48	0.48
No cluster	RandomForest	0.62	0.62	0.62	0.62
No cluster	XGBoost	0.63	0.63	0.63	0.63
Cluster 1	Multinomial Logistic Regression	0.58	0.57	0.58	0.57
Cluster 1	CART	0.49	0.50	0.49	0.49
Cluster 1	RandomForest	0.60	0.60	0.60	0.60
Cluster 1	XGBoost	0.63	0.63	0.63	0.63
Cluster 2	Multinomial Logistic Regression	0.53	0.52	0.53	0.51
Cluster 2	CART	0.49	0.50	0.49	0.49
Cluster 2	RandomForest	0.61	0.62	0.61	0.61
Cluster 2	XGBoost	0.62	0.62	0.62	0.62

Table 6 indicates that clustering does not materially enhance predictive performance for TCRI classification. Multinomial logistic regression shows only a modest improvement within Cluster 1 (0.58 versus 0.54 without clustering), but the gains are inconsistent across clusters and negligible for ensemble methods such as Random Forest and XGBoost, which remain stable around 0.60-0.63 regardless of segmentation. CART continues to underperform in all settings. Given these outcomes, the additional complexity introduced by clustering is not justified by measurable accuracy improvements. Eliminating clustering from the prediction pipeline streamlines the modeling process and highlights that ensemble methods, particularly XGBoost, are sufficiently robust to capture heterogeneity in firm-level financial data without prior segmentation. This finding underscores that clustering may serve interpretive purposes in subgroup analysis but is not necessary for predictive accuracy in the context of TCRI.

5. Conclusion

This study compared hybrid and standalone machine learning models to determine the best architecture for corporate credit risk assessment, directly addressing the limitations of linear statistical methods in handling heterogeneous and nonlinear financial data. Using the Taiwan Corporate Credit Risk Index (TCRI) dataset, three major findings emerged. First, XGBoost consistently achieved the highest predictive accuracy (0.63), confirming the importance of ensemble methods for capturing complex financial relationships. Second, the cluster-then-predict strategy with K-means segmentation did not improve accuracy, showing that ensemble models are robust enough to manage data heterogeneity without the added computational burden of pre-clustering. Third, backward feature selection distilled the dataset into ten economically meaningful predictors, balancing statistical rigor with interpretability.

Although accuracy levels remain modest, the machine learning pipeline demonstrated consistent improvements over traditional approaches, establishing a solid methodological foundation for future research. The results advocate for a streamlined, highperformance modeling pipeline that prioritizes ensemble methods and efficient feature engineering, while strategically rejecting preclustering. The most significant policy implication is the tradeoff between model complexity and predictive gain: adopting a simpler yet highly accurate XGBoostbased framework benefits financial institutions by optimizing resources and regulators by ensuring transparency and stability in credit risk models.

Future work should explore alternative segmentation methods such as Hierarchical Clustering and DBSCAN to test whether more structure-preserving or outlier-robust approaches can yield homogeneous subgroups. At the same time, new benchmark classifiers like Generalized Additive Models (GAMs) and basic Neural Networks (Multi-Layer Perceptrons) should be evaluated to balance nonlinearity capture with interpretability. Only after exhausting these foundational techniques should research advance to more sophisticated ensembles (LightGBM, CatBoost) and integrate Explainable AI tools like SHAP to meet regulatory and institutional demands for transparency in credit rating systems.

Appendix A.

Table A.7: Financial predictors and data types

No.	Predictors	Type
1	Cash and cash equivalents	Continuous
2	Current assets	Continuous
3	Total assets	Continuous
4	Current liabilities	Continuous
5	Total liabilities	Continuous
6	Common shares	Continuous
7	Share capital	Continuous
8	Total capital	Continuous
9	Retained earnings	Continuous
10	Total shareholder's equity	Continuous
11	Net operating income	Continuous
12	COGS	Continuous
13	Operating income	Continuous
14	Operating expense	Continuous
15	Working capital	Continuous
16	R&D expense	Continuous
17	Expected credit impairment (loss) profit - operating expense	Continuous
18	Operating profit	Continuous
19	Finance costs	Continuous
20	Net income before tax	Continuous
21	Net income before interest and tax	Continuous
22	Cash flow from operations	Continuous
23	Cash flow from investing activities	Continuous
24	Cash flow from financing activities	Continuous
25	Long-term and short-term borrowings	Continuous
26	Net liabilities	Continuous
27	Tobin's Q	Continuous
28	Financial leverage	Continuous
29	Operating leverage	Continuous
30	Interest coverage ratio	Continuous
31	ROA - comprehensive income	Continuous

Continued on next page

No.	Predictors	Type
32	ROE - comprehensive income	Continuous
33	Cash flow adequacy ratio	Continuous
34	Cash reinvestment ratio	Continuous
35	Current ratio	Continuous
36	Quick ratio	Continuous
37	Interest expense ratio	Continuous
38	Interest expense ratio (B)	Continuous
39	Total liabilities / total net worth	Continuous
40	Debt ratio	Continuous
41	Net worth / assets	Continuous
42	Long-term funding ratio (A)	Continuous
43	Debt dependency ratio	Continuous
44	Contingent liabilities to net worth	Continuous
45	Operating profit / paid-in capital ratio	Continuous
46	Pre-tax net profit / paid-in capital ratio	Continuous
47	Retention ratio	Continuous
48	Dividend payout ratio	Continuous

Appendix B.

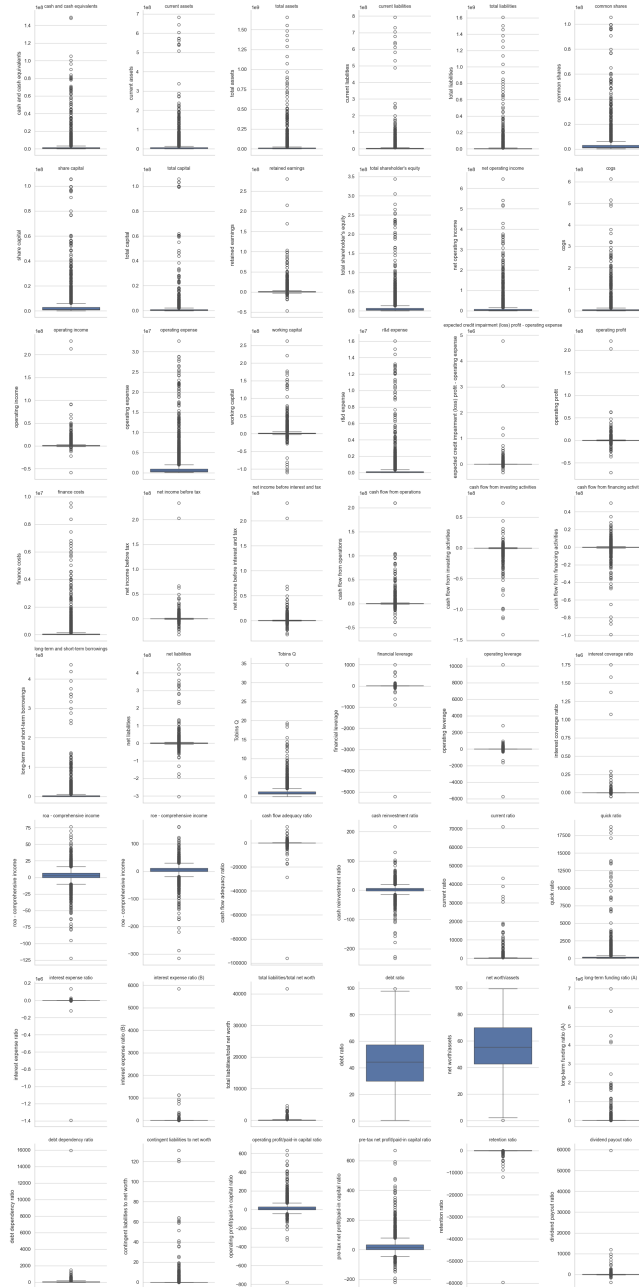


Figure B.3: Box plot

Table B.8: Outlier summary of financial metrics

Metric	Count	Percentage (%)
Operating expense	130	1.96
ROA comprehensive income	117	1.77
Common shares	114	1.72
Share capital	114	1.72
ROE comprehensive income	111	1.68
Cash reinvestment ratio	109	1.65
Net operating income	108	1.63
COGS	107	1.62
Pre-tax net profit / paid-in capital	103	1.56
Tobins Q	96	1.45
Operating profit / paid-in capital	95	1.43
Total shareholders equity	94	1.42
Finance costs	87	1.31
Cash and cash equivalents	86	1.30
Total assets	80	1.21
Retained earnings	78	1.18
Working capital	78	1.18
Cash flow from financing activities	77	1.16
Current assets	75	1.13
Borrowings (long + short)	70	1.06
R&D expense	69	1.04
Cash flow from operations	69	1.04
Operating income	68	1.03
Cash flow from investing activities	67	1.01
Total liabilities	57	0.86
Total capital	54	0.82
Net liabilities	54	0.82
Current liabilities	51	0.77
Quick ratio	50	0.76
Operating profit	45	0.68
Net income before interest and tax	41	0.62
Net income before tax	38	0.57
Long-term funding ratio (A)	38	0.57
Current ratio	36	0.54
Contingent liabilities / net worth	35	0.53
Expected credit impairment	25	0.38

Continued on next page

Metric	Count	Percentage (%)
Cash flow adequacy ratio	21	0.32
Total liabilities / net worth	16	0.24
Operating leverage	12	0.18
Dividend payout ratio	12	0.18
Retention ratio	11	0.17
Interest coverage ratio	11	0.17
Debt dependency ratio	10	0.15
Interest expense ratio (B)	10	0.15
Financial leverage	9	0.14
Interest expense ratio	3	0.05
Debt ratio	0	0.00
Net worth / assets	0	0.00

Appendix C.

Table C.9: Descriptive statistics of financial predictors

Predictor	Count	Mean	Std	Min	25%	50%	75%	Max	Skewness	Kurtosis
cash and cash equivalents	6622	2031160.22	6681963.97	1125.00	319784.25	701200.50	1517499.50	149427959.00	10.50	145.28
current assets	6622	8678525.12	29277931.84	15260.00	1446458.50	2779874.00	6564517.25	683292033.00	13.36	235.91
total assets	6622	18776722.90	71559079.54	17267.00	2764087.75	5434440.50	12432489.75	1657801534.00	12.45	204.25
current liabilities	6622	5896663.70	28378831.46	776.00	653886.00	1491492.00	3771463.25	793421493.00	18.69	421.01
total liabilities	6622	11552982.53	62382019.86	4946.00	944673.75	2295347.00	5886414.00	1607622209.00	15.82	307.62
common shares	6622	3387926.46	7639833.08	142719.00	871349.50	1489517.50	3005155.50	105596201.00	7.77	77.31
share capital	6622	3400313.46	7710776.71	157600.00	875460.00	1490550.00	3005579.00	105596201.00	7.85	78.69
total capital	6622	1191068.29	5305519.35	-207088.00	59116.25	305537.50	879875.25	105919710.00	14.13	236.68
retained earnings	6622	2141344.90	7689971.45	-46658745.00	187388.25	724805.00	1799004.25	280077242.00	15.65	413.02
total shareholder's equity	6622	7223740.37	18046458.62	2927.00	1610381.25	2993183.50	6311369.00	343785963.00	9.12	109.64
net operating income	6622	9697199.45	29934146.98	11.00	1451971.00	3112933.00	7313326.50	646261954.00	9.47	121.26
cogs	6622	8172317.26	26742182.96	-4032.00	1031778.75	2421068.00	5934386.75	612924800.00	9.85	135.27
operating income	6622	1524882.19	5450723.43	-58604462.00	251877.75	610972.50	1317541.75	230170119.00	23.12	847.75
operating expense	6622	1046714.74	2289484.96	7909.00	226361.25	426035.50	937637.25	32681953.00	6.51	54.79
working capital	6622	2781861.43	9915462.68	-110129460.00	550485.25	1234237.00	2585429.50	262648808.00	10.24	207.37
r&d expense	6622	222928.62	899669.89	0.00	3993.75	50964.50	143907.75	16028319.00	10.70	135.08
expected credit impairment (loss) profit - operating expense	6622	4643.18	77871.40	-326032.00	0.00	0.00	60.75	4778808.00	45.51	2503.25
operating profit	6622	479790.76	4534595.66	-71751974.00	-17919.25	142005.00	454621.75	220698167.00	31.62	1472.26
finance costs	6622	109532.60	444821.87	0.00	4889.25	18172.50	57624.00	9556030.00	11.00	157.61
net income before tax	6622	578370.24	4533243.44	-31838025.00	18272.75	198254.00	545032.50	233620714.00	35.47	1666.37
net income before interest and tax	6622	687902.84	4648289.44	-27876062.00	37419.25	225363.50	604346.75	236018505.00	34.21	1573.98
cash flow from operations	6622	906186.26	6459098.41	-64409953.00	1467.75	216161.50	647191.75	210067260.00	16.49	414.29
cash flow from investing activities	6622	-675145.01	4779710.20	-141108194.00	-441924.75	-125219.50	-5005.00	73301769.00	-14.27	332.43
cash flow from financing activities	6622	-141479.17	3552694.69	-99260726.00	-315319.50	-59028.00	170383.00	50257136.00	-11.55	280.33
long-term and short-term borrowings	6622	5183906.95	19378873.84	0.00	237297.75	1057886.00	3352574.00	448566805.00	12.63	214.39

Continued on next page

Predictor	Count	Mean	Std	Min	25%	50%	75%	Max	Skewness	Kurtosis
net liabilities	6622	2563660.09	18088300.10	-303180654.00	-620917.00	153718.00	1800306.00	446539743.00	12.33	274.11
Tobins Q	6622	1.18	1.12	0.02	0.72	0.92	1.29	34.68	9.59	178.62
financial leverage	6622	0.43	68.14	-5215.02	1.00	1.03	1.11	998.31	-67.44	5200.70
operating leverage	6622	5.11	153.59	-5716.41	1.15	1.89	3.41	10160.76	36.64	3198.65
interest coverage ratio	6622	1392.41	36867.99	-52212.75	1.94	9.84	32.14	1750954.00	39.83	1677.22
roa - comprehensive income	6622	2.94	8.78	-122.36	-0.08	2.92	6.46	76.71	-1.27	20.99
roe - comprehensive income	6622	4.91	18.33	-315.90	-0.17	5.55	12.19	162.46	-3.02	43.53
cash flow adequacy ratio	6622	36.40	1357.38	-96151.45	24.36	71.61	114.55	13547.99	-55.84	3852.28
cash reinvestment ratio	6622	1.89	13.61	-232.86	-1.90	2.38	6.75	216.95	-2.24	54.87
current ratio	6622	359.32	1518.88	5.88	135.68	183.70	285.42	71149.48	27.09	973.15
quick ratio	6622	238.54	789.13	0.43	68.28	114.36	197.59	18744.78	14.63	262.14
interest expense ratio	6622	-190.02	17271.57	-1393110.22	0.04	4.24	15.27	137412.37	-79.22	6393.04
interest expense ratio (B)	6622	3.67	77.25	0.00	0.16	0.56	1.36	5851.14	67.10	4980.85
total liabilities/total net worth	6622	120.83	543.07	0.48	42.78	80.32	134.23	41648.92	68.02	5168.92
debt ratio	6622	43.72	19.08	0.48	29.96	44.54	57.31	99.76	0.01	-0.52
net worth/assets	6622	56.28	19.08	0.24	42.69	55.46	70.04	99.52	-0.01	-0.52
long-term funding ratio (A)	6622	13820.50	173544.77	0.00	173.57	276.58	544.22	7004220.00	25.00	778.50
debt dependency ratio	6622	61.93	210.78	0.00	10.84	39.84	79.94	15944.99	64.90	4872.20
contingent liabilities to net worth	6622	0.29	3.61	0.00	0.00	0.00	0.00	131.01	24.42	725.44
operating profit/paid-in capital ratio	6622	17.17	40.67	-777.80	-1.21	9.09	28.01	631.99	2.51	58.18
pre-tax net profit/paid-in capital ratio	6622	20.91	41.80	-219.72	1.33	12.77	32.40	669.00	3.96	37.38
retention ratio	6622	15.42	776.76	-59557.14	21.86	37.28	54.19	100.00	-69.23	5240.15
dividend payout ratio	6622	68.59	787.35	-3700.15	0.00	50.16	78.41	59657.14	66.77	4972.56

Declaration

Declaration of Pedagogical Purpose

This manuscript is prepared as part of the course project for Machine Learning and FinTech (Fall 2025) at National Yang Ming Chiao Tung University, instructed by Professor Huei-Wen Teng. The work is intended solely for pedagogical purposes, including learning, discussion, and academic training. It does not represent original research submitted for journal publication, nor should it be considered as professional financial advice.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used [ChatGPT] and [Grok], developed by [OpenAI] and [xAI], respectively, in order to improve the clarity, precision, and overall quality of the writing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- Afik, Z., Galil, K., 2024. Have ratings become more accurate? *Journal of Banking & Finance* 155, 107337. doi:10.1016/j.jbankfin.2024.107337.
- Barboza, F., Kimura, H., Altman, E., 2017. Machine learning models and bankruptcy prediction. *Expert systems with applications* 83, 405–417.
- Galil, K., Hauptman, A., Levy Rosenboim, R., 2023. Prediction of corporate credit ratings with machine learning: Simple interpretative models. *SSRN Electronic Journal* doi:10.2139/ssrn.4504054. preprint, not peer reviewed.
- Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., Pedreschi, D., 2018. A survey of methods for explaining black box models. *ACM computing surveys (CSUR)* 51, 1–42.
- Hajek, P., Sahut, J.M., Olej, V., 2024. Credit rating prediction using a fuzzy mcdm approach with criteria interactions and topsis sorting. *Annals of Operations Research* doi:10.1007/s10479-024-06183-2.
- Hsieh, N.C., 2005. Hybrid mining approach in the design of credit scoring models. *Expert Systems with Applications* 28, 655–665. doi:10.1016/j.eswa.2004.12.022.

- Teng, H.W., Kang, M.H., Lee, I.H., Bai, L.C., 2024. Bridging accuracy and interpretability: A rescaled cluster-then-predict approach for enhanced credit scoring. *International Review of Financial Analysis* 91, 103005. doi:10.1016/j.irfa.2023.103005.
- Tsai, C.F., Chen, M.L., 2010. Credit rating by hybrid machine learning techniques. *Applied Soft Computing* 10, 374–380. doi:10.1016/j.asoc.2009.08.003.
- Wu, H.C., Hu, Y.H., Huang, Y.H., 2014. Two-stage credit rating prediction using machine learning techniques. *Kybernetes* 43, 1098–1113. doi:10.1108/K-10-2013-0218.